

tent_count

```
library(ggplot2)
library(ggpubr)
library(plotrix)
library(tidyverse)
library(dplyr)
library(car)
library(lme4)
library(emmeans)
library(qqplotr)
library(here)
library(scales)
library(janitor)
library(gt)
library(gh4x)
```

```
rm(list = ls())
#graphics.off()
```

```
getwd()
```

```
## [1] "/Users/junbc/Downloads"
```

```
setwd("~/Documents/GitHub/heating_lacerates_final")
```

```
select <- dplyr::select
filter <- dplyr::filter
mutate <- dplyr::mutate
lag <- dplyr::lag
```

```
## Experiment 2022
```

```
tentcount_2022 <- read.csv("~/Documents/GitHub/heating_lacerates_final/data/2022_tentcount.csv")
```

```
str(tentcount_2022)
```

```
## 'data.frame': 4635 obs. of 10 variables:
## $ ID : chr "CC7-AP0-1_B3.2" "CC7-AP0-1_B3.2" "CC7-AP0-1_B3.2" "CC7-AP0-1_B3.2" ...
## $ plate : chr "CC7-AP0-1" "CC7-AP0-1" "CC7-AP0-1" "CC7-AP0-1" ...
## $ treatment : chr "CC7-AP0-25C" "CC7-AP0-25C" "CC7-AP0-25C" "CC7-AP0-25C" ...
## $ line : chr "CC7" "CC7" "CC7" "CC7" ...
## $ symbiosis : chr "Apo" "Apo" "Apo" "Apo" ...
## $ well : chr "B3" "B3" "B3" "B3" ...
## $ temp : chr "25C (ambient)" "25C (ambient)" "25C (ambient)" "25C (ambient)" ...
## $ day : int 0 4 5 6 7 8 9 10 11 12 ...
```

```
## $ day_cat : chr "day_0" "day_4" "day_5" "day_6" ...
## $ tent_count: chr "0" "0" "0" "0" ...
```

```
tentcount_2022$tent_count <- as.numeric(tentcount_2022$tent_count)
tentcount_2022$ID <- as.factor(tentcount_2022$ID)
tentcount_2022$plate <- as.factor(tentcount_2022$plate)
tentcount_2022$well <- as.factor(tentcount_2022$well)
tentcount_2022$line <- as.factor(tentcount_2022$line)
tentcount_2022$temp <- as.factor(tentcount_2022$temp)
tentcount_2022$treatment <- as.factor(tentcount_2022$treatment)
tentcount_2022$symbiosis <- as.factor(tentcount_2022$symbiosis)
tentcount_2022$day <- as.numeric(tentcount_2022$day)
tentcount_2022$day_cat <- as.factor(tentcount_2022$day_cat)

tentcount_2022 <- tentcount_2022 %>%
  mutate(Day = as.factor(day)) %>%
  mutate(Day = dplyr::recode(Day, "0" = "00"))

tentcount_2022_cleaned <- tentcount_2022

data_means <- tentcount_2022_cleaned %>%
  group_by(treatment, day) %>%
  summarise(mean = mean(tent_count, na.rm=TRUE),
            se = std.error(tent_count, na.rm=TRUE))

All_2022_plot <- ggplot(data = data_means, aes(x = day, y = mean)) +
  theme_classic(base_size = 15) +
  geom_line(aes(color = treatment, group = treatment),
            position = position_dodge(0.5)) +
  ylab(bquote("Mean tentacle number")) +
  xlab("Days post laceration (dpl)") +
  ggtitle("Effect of Temperature on Pedal Lacerate Tentacle Development in Aiptasia") +
  ylim(0,12) +
  geom_point(aes(color = treatment),
             size = 2.5,
             shape = 20,
             position = position_dodge(0.5)) +
  scale_x_continuous(
    breaks = round(seq(min(data_means$day), max(data_means$day), by = 1), 1)
  ) +
  geom_errorbar(
    aes(color = treatment, ymin = mean - se, ymax = mean + se),
    width = 0.2,
    position = position_dodge(0.5)
  ) +
  scale_color_manual(
    values = c(
      "CC7-AP0-25C" = "aquamarine",
      "CC7-AP0-32C" = "chocolate",
      "CC7-SYM-25C" = "darkorchid",
      "CC7-SYM-32C" = "coral1",
      "H2-AP0-25C" = "cornflowerblue",
      "H2-AP0-32C" = "orange",
      "H2-SYM-25C" = "blue",
    )
  )
```

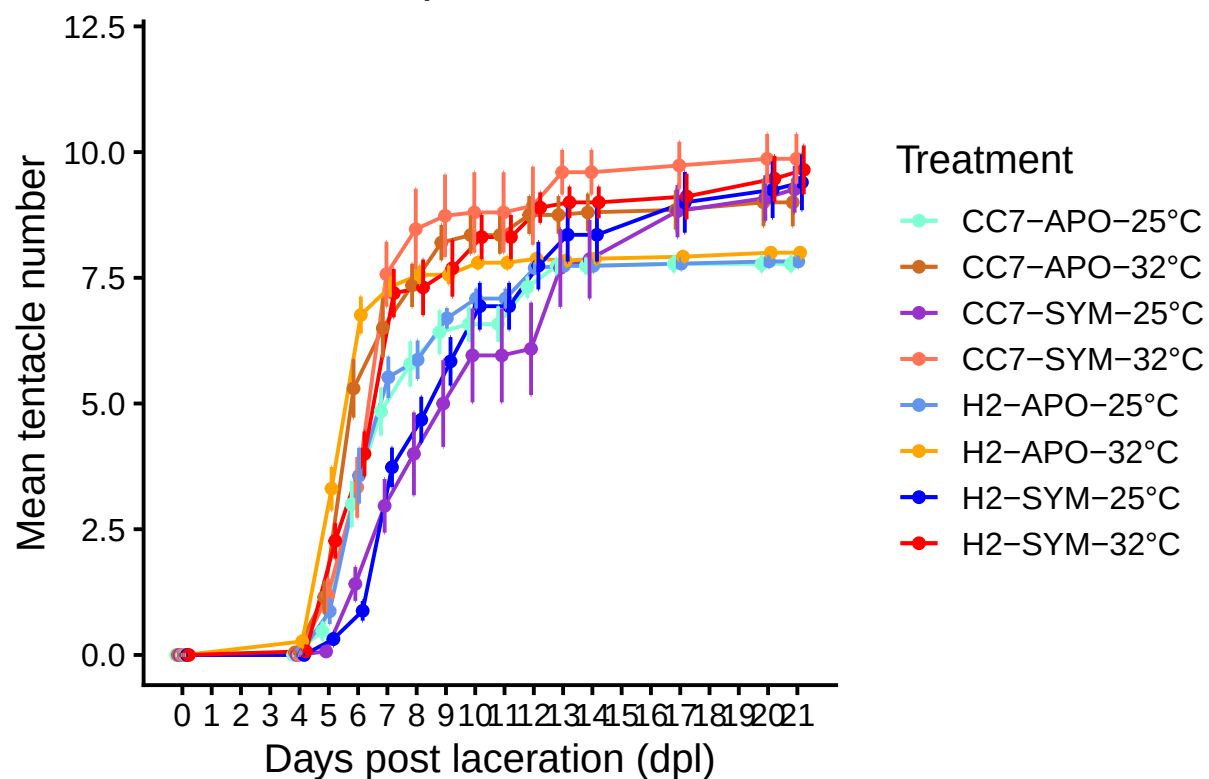
```

"H2-SYM-32C" = "red"
),
breaks = c(
  "CC7-APO-25C", "CC7-APO-32C",
  "CC7-SYM-25C", "CC7-SYM-32C",
  "H2-APO-25C", "H2-APO-32C",
  "H2-SYM-25C", "H2-SYM-32C"
),
labels = c(
  "CC7-APO-25°C",
  "CC7-APO-32°C",
  "CC7-SYM-25°C",
  "CC7-SYM-32°C",
  "H2-APO-25°C",
  "H2-APO-32°C",
  "H2-SYM-25°C",
  "H2-SYM-32°C"
)
) +
theme(legend.text.align = 0) +
scale_size_manual(values = c(1.2,1.2,1.2,1.2)) +
labs(colour = "Treatment")

```

All_2022_plot

Effect of Temperature on Pedal Lacerate Tentacle D



```
## Tentacle count analysis
```

```
library(lme4)
library(car)
library(emmeans)
library(performance)
library(DHARMA)
```

```
data <- tentcount_2022_cleaned %>%
  filter(line != "CC7") %>%
  mutate(
    day_cat = factor(day_cat),
    temp = factor(temp),
    symbiosis = factor(symbiosis),
    ID = factor(ID)
  )
```

```
data <- tentcount_2022_cleaned
str(data)
```

```
## 'data.frame': 4635 obs. of 11 variables:
## $ ID : Factor w/ 309 levels "CC7-APO-1_A1",...: 8 8 8 8 8 8 8 8 8 ...
## $ plate : Factor w/ 24 levels "CC7-APO-1","CC7-APO-2",...: 1 1 1 1 1 1 1 1 1 ...
## $ treatment : Factor w/ 8 levels "CC7-APO-25C",...: 1 1 1 1 1 1 1 1 ...
## $ line : Factor w/ 2 levels "CC7","H2": 1 1 1 1 1 1 1 1 ...
## $ symbiosis : Factor w/ 2 levels "Apo","Sym": 1 1 1 1 1 1 1 1 ...
## $ well : Factor w/ 12 levels "A1","A2","A3",...: 7 7 7 7 7 7 7 7 ...
## $ temp : Factor w/ 2 levels "25C (ambient)",...: 1 1 1 1 1 1 1 1 ...
## $ day : num 0 4 5 6 7 8 9 10 11 12 ...
## $ day_cat : Factor w/ 15 levels "day_0","day_10",...: 1 10 11 12 13 14 15 2 3 4 ...
## $ tent_count: num 0 0 0 0 1 3 5 5 5 6 ...
## $ Day : Factor w/ 15 levels "00","4","5","6",...: 1 2 3 4 5 6 7 8 9 10 ...
```

```
model_pois <- glmer(
  tent_count ~ temp * symbiosis * day_cat + (1 | ID),
  family = poisson(link = "log"),
  data = data,
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)
```

```
summary(model_pois)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: poisson ( log )
## Formula: tent_count ~ temp * symbiosis * day_cat + (1 | ID)
## Data: data
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
## AIC BIC logLik -2*log(L) df.resid
```

```

##    10635.5    11001.2    -5256.7    10513.5    2904
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.6825 -0.2409  0.0000  0.2174  9.5187
##
## Random effects:
##   Groups Name   Variance Std.Dev.
##   ID      (Intercept) 0.07849  0.2802
## Number of obs: 2965, groups: ID, 309
##
## Fixed effects:
##                                     Estimate Std. Error z value
## (Intercept)                       -2.031e+01  3.512e+03  -0.006
## temp32C (heat stress)                8.221e-03  5.004e+03   0.000
## symbiosisSym                        1.121e-02  4.320e+03   0.000
## day_catday_10                       2.222e+01  3.512e+03   0.006
## day_catday_11                       2.222e+01  3.512e+03   0.006
## day_catday_12                       2.231e+01  3.512e+03   0.006
## day_catday_13                       2.234e+01  3.512e+03   0.006
## day_catday_14                       2.234e+01  3.512e+03   0.006
## day_catday_17                       2.234e+01  3.512e+03   0.006
## day_catday_20                       2.235e+01  3.512e+03   0.006
## day_catday_21                       2.235e+01  3.512e+03   0.006
## day_catday_4                        1.656e+01  3.512e+03   0.005
## day_catday_5                        1.992e+01  3.512e+03   0.006
## day_catday_6                        2.149e+01  3.512e+03   0.006
## day_catday_7                        2.195e+01  3.512e+03   0.006
## day_catday_8                        2.206e+01  3.512e+03   0.006
## day_catday_9                        2.218e+01  3.512e+03   0.006
## temp32C (heat stress):symbiosisSym  -2.764e-03  6.112e+03   0.000
## temp32C (heat stress):day_catday_10  1.501e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_11  1.501e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_12  8.454e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_13  5.376e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_14  5.914e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_17  6.353e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_20  7.365e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_21  7.365e-02  5.004e+03   0.000
## temp32C (heat stress):day_catday_4   1.981e+00  5.004e+03   0.000
## temp32C (heat stress):day_catday_5   1.225e+00  5.004e+03   0.000
## temp32C (heat stress):day_catday_6   6.037e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_7   2.785e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_8   2.372e-01  5.004e+03   0.000
## temp32C (heat stress):day_catday_9   1.675e-01  5.004e+03   0.000
## symbiosisSym:day_catday_10          -1.017e-01  4.320e+03   0.000
## symbiosisSym:day_catday_11          -1.017e-01  4.320e+03   0.000
## symbiosisSym:day_catday_12          -1.179e-01  4.320e+03   0.000
## symbiosisSym:day_catday_13          -8.517e-03  4.320e+03   0.000
## symbiosisSym:day_catday_14           6.218e-04  4.320e+03   0.000
## symbiosisSym:day_catday_17           1.043e-01  4.320e+03   0.000
## symbiosisSym:day_catday_20           1.295e-01  4.320e+03   0.000
## symbiosisSym:day_catday_21           1.471e-01  4.320e+03   0.000
## symbiosisSym:day_catday_4          -1.655e+01  5.236e+03  -0.003

```

## symbiosisSym:day_catday_5	-1.233e+00	4.320e+03	0.000
## symbiosisSym:day_catday_6	-1.164e+00	4.320e+03	0.000
## symbiosisSym:day_catday_7	-4.862e-01	4.320e+03	0.000
## symbiosisSym:day_catday_8	-3.355e-01	4.320e+03	0.000
## symbiosisSym:day_catday_9	-2.324e-01	4.320e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_10	1.400e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_11	1.400e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_12	1.787e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_13	1.046e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_14	9.008e-02	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_17	-1.014e-02	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_20	-2.234e-02	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_21	-3.029e-02	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_4	1.524e+01	6.791e+03	0.002
## temp32C (heat stress):symbiosisSym:day_catday_5	9.785e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_6	6.491e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_7	5.043e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_8	3.528e-01	6.112e+03	0.000
## temp32C (heat stress):symbiosisSym:day_catday_9	2.436e-01	6.112e+03	0.000
##	Pr(> z)		
## (Intercept)	0.995		
## temp32C (heat stress)	1.000		
## symbiosisSym	1.000		
## day_catday_10	0.995		
## day_catday_11	0.995		
## day_catday_12	0.995		
## day_catday_13	0.995		
## day_catday_14	0.995		
## day_catday_17	0.995		
## day_catday_20	0.995		
## day_catday_21	0.995		
## day_catday_4	0.996		
## day_catday_5	0.995		
## day_catday_6	0.995		
## day_catday_7	0.995		
## day_catday_8	0.995		
## day_catday_9	0.995		
## temp32C (heat stress):symbiosisSym	1.000		
## temp32C (heat stress):day_catday_10	1.000		
## temp32C (heat stress):day_catday_11	1.000		
## temp32C (heat stress):day_catday_12	1.000		
## temp32C (heat stress):day_catday_13	1.000		
## temp32C (heat stress):day_catday_14	1.000		
## temp32C (heat stress):day_catday_17	1.000		
## temp32C (heat stress):day_catday_20	1.000		
## temp32C (heat stress):day_catday_21	1.000		
## temp32C (heat stress):day_catday_4	1.000		
## temp32C (heat stress):day_catday_5	1.000		
## temp32C (heat stress):day_catday_6	1.000		
## temp32C (heat stress):day_catday_7	1.000		
## temp32C (heat stress):day_catday_8	1.000		
## temp32C (heat stress):day_catday_9	1.000		
## symbiosisSym:day_catday_10	1.000		
## symbiosisSym:day_catday_11	1.000		

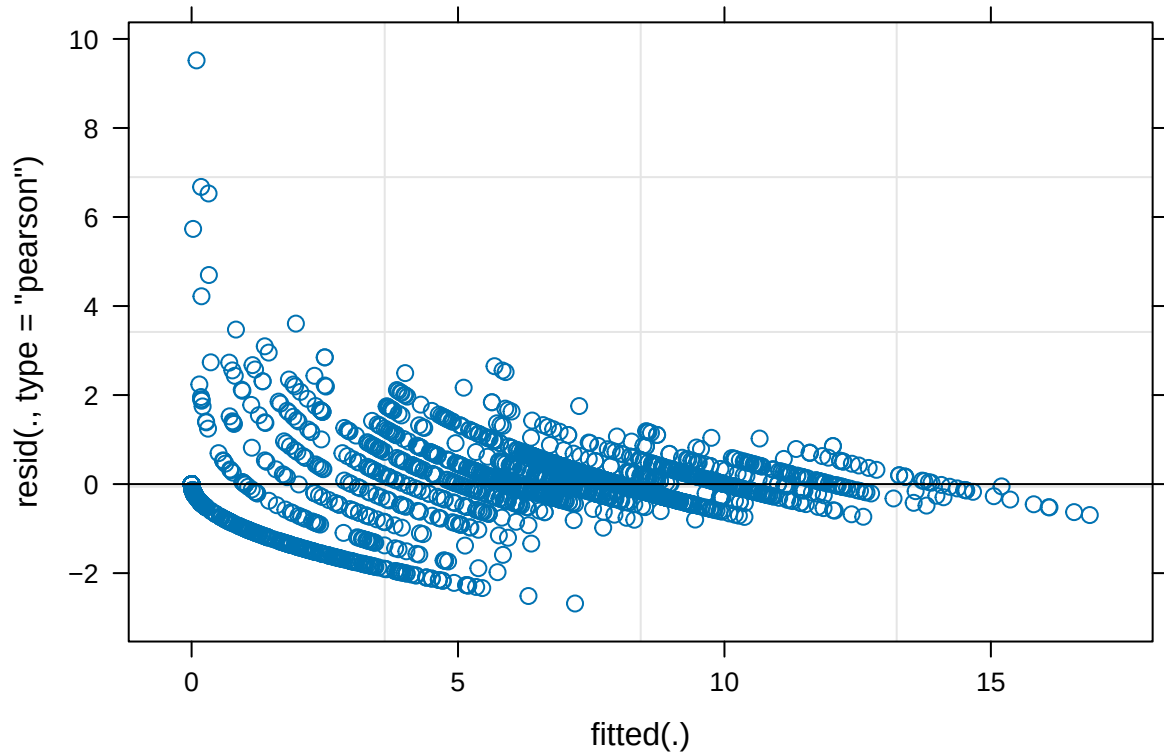
```

## symbiosisSym:day_catday_12      1.000
## symbiosisSym:day_catday_13      1.000
## symbiosisSym:day_catday_14      1.000
## symbiosisSym:day_catday_17      1.000
## symbiosisSym:day_catday_20      1.000
## symbiosisSym:day_catday_21      1.000
## symbiosisSym:day_catday_4       0.997
## symbiosisSym:day_catday_5       1.000
## symbiosisSym:day_catday_6       1.000
## symbiosisSym:day_catday_7       1.000
## symbiosisSym:day_catday_8       1.000
## symbiosisSym:day_catday_9       1.000
## temp32C (heat stress):symbiosisSym:day_catday_10 1.000
## temp32C (heat stress):symbiosisSym:day_catday_11 1.000
## temp32C (heat stress):symbiosisSym:day_catday_12 1.000
## temp32C (heat stress):symbiosisSym:day_catday_13 1.000
## temp32C (heat stress):symbiosisSym:day_catday_14 1.000
## temp32C (heat stress):symbiosisSym:day_catday_17 1.000
## temp32C (heat stress):symbiosisSym:day_catday_20 1.000
## temp32C (heat stress):symbiosisSym:day_catday_21 1.000
## temp32C (heat stress):symbiosisSym:day_catday_4  0.998
## temp32C (heat stress):symbiosisSym:day_catday_5  1.000
## temp32C (heat stress):symbiosisSym:day_catday_6  1.000
## temp32C (heat stress):symbiosisSym:day_catday_7  1.000
## temp32C (heat stress):symbiosisSym:day_catday_8  1.000
## temp32C (heat stress):symbiosisSym:day_catday_9  1.000

## optimizer (bobyqa) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate Hessian with 4 negative eigenvalues

```

```
plot(model_pois)
```

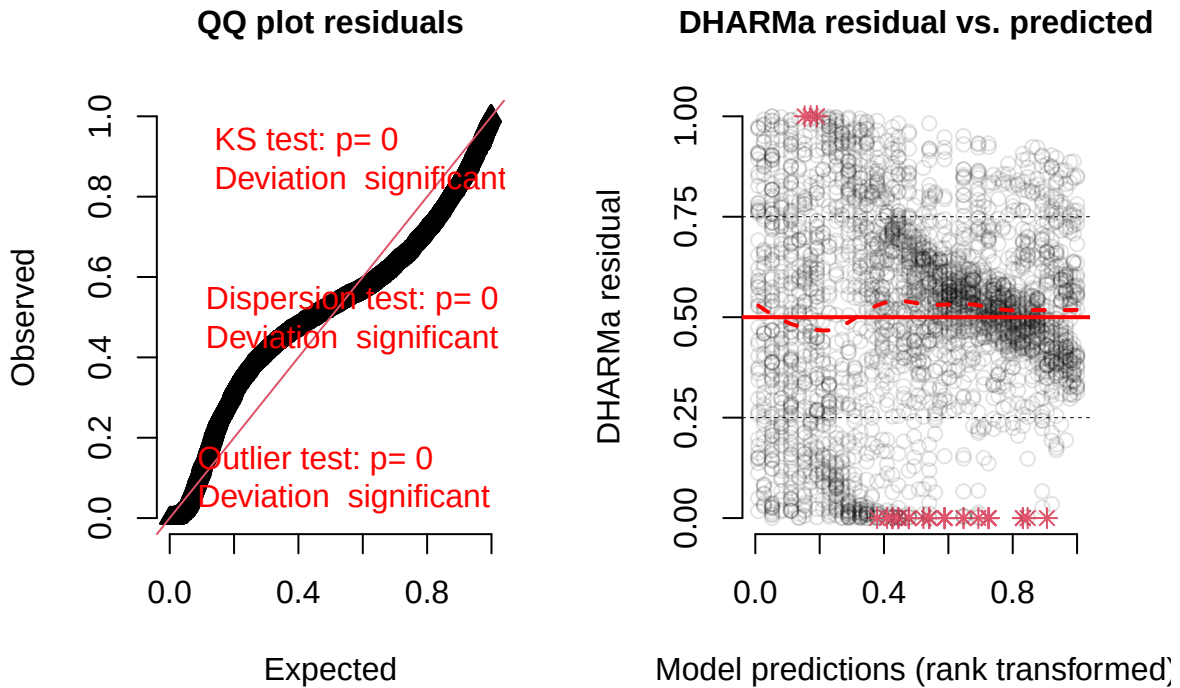


```
check_overdispersion(model_pois)
```

```
## # Overdispersion test
##
##      dispersion ratio =    0.529
##   Pearson's Chi-Squared = 1537.053
##              p-value =      1
```

```
sim_res <- simulateResiduals(model_pois)
plot(sim_res)
```


DHARMA residual



```
anova_tentacle <- Anova(model_pois, type = "II")
```

```
anova_tentacle
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: tent_count
##
```

	Chisq	Df	Pr(>Chisq)
temp	287.089	27	< 2.2e-16 ***
symbiosis	173.768	27	< 2.2e-16 ***
day_cat	1801.773	52	< 2.2e-16 ***
temp:symbiosis	38.127	13	0.0002753 ***
temp:day_cat	236.544	26	< 2.2e-16 ***
symbiosis:day_cat	170.146	26	< 2.2e-16 ***
temp:symbiosis:day_cat	33.365	14	0.0025514 **

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova_tentacle_df <- as.data.frame(anova_tentacle) %>%
  tibble::rownames_to_column("Factor")
```

```
write.csv(
  anova_tentacle_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
```

```

    "TableS1_anova_tentacle_number.csv"
  ),
  row.names = FALSE
)

emm_temp <- emmeans(model_pois, ~ temp | day_cat, type = "response")
tukey_temp <- pairs(emm_temp, adjust = "tukey")

tukey_temp_df <- as.data.frame(tukey_temp)

write.csv(
  tukey_temp_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS2_tukey_temperature_tentacle_number.csv"
  ),
  row.names = FALSE
)

emm_symbiosis <- emmeans(model_pois, ~ symbiosis | day_cat, type = "response")
tukey_sym <- pairs(emm_symbiosis, adjust = "tukey")

tukey_sym_df <- as.data.frame(tukey_sym)

fig1_dat_raw <- tentcount_2022_cleaned %>%
  filter(line != "CC7") %>%
  mutate(
    day_cat = factor(day_cat),
    temp = factor(temp),
    symbiosis = factor(symbiosis),
    ID = factor(ID)
  )

fig1fuller_data <- fig1_dat_raw %>%
  group_by(temp, symbiosis, day) %>%
  summarise(
    mean = mean(tent_count, na.rm = TRUE),
    se = std.error(tent_count, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    treatment = case_when(
      symbiosis == "Apo" & temp == "25C (ambient)" ~ "Apo, 25°C",
      symbiosis == "Sym" & temp == "25C (ambient)" ~ "Sym, 25°C",
      symbiosis == "Apo" & temp == "32C (heat stress)" ~ "Apo, 32°C",
      symbiosis == "Sym" & temp == "32C (heat stress)" ~ "Sym, 32°C"
    ),
    treatment = factor(
      treatment,
      levels = c("Apo, 25°C", "Sym, 25°C", "Apo, 32°C", "Sym, 32°C")
    )
  )

```

```

my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

fig1_fuller <- ggplot(
  fig1fuller_data,
  aes(
    x = day,
    y = mean,
    color = treatment,
    linetype = treatment,
    group = treatment
  )
) +
  my_theme +
  geom_line(linewidth = 1.2) +
  geom_errorbar(
    aes(ymin = mean - se, ymax = mean + se),
    width = 0.2,
    linewidth = 0.6,
    alpha = 0.8,
    linetype = "solid"
  ) +
  geom_point(size = 4) +

  annotate("segment", x = 5, xend = 11, y = 9.2, yend = 9.2, linewidth = 0.8) +
  annotate("text", x = 8, y = 9.45, label = "***", size = 6) +
  annotate("text", x = 12, y = 9.45, label = "***", size = 6) +

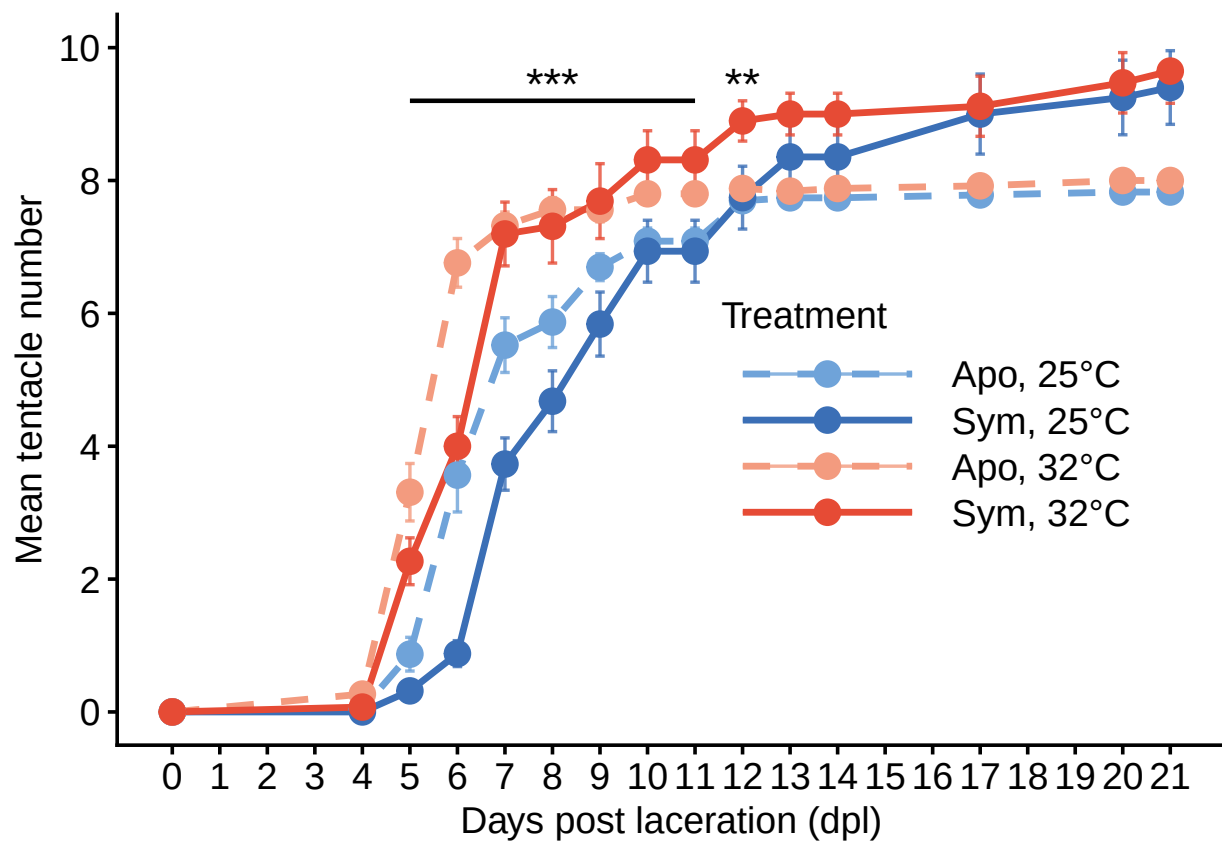
  ylab("Mean tentacle number") +
  xlab("Days post laceration (dpl)") +
  scale_y_continuous(
    breaks = seq(0, 10, 2),
    limits = c(0, 10)
  ) +
  scale_x_continuous(
    breaks = seq(min(fig1fuller_data$day), max(fig1fuller_data$day), 1)
  ) +
  scale_color_manual(
    values = c(
      "Apo, 25°C" = "#6FA3D9",
      "Sym, 25°C" = "#3B6FB6",
      "Apo, 32°C" = "#F39B7F",
      "Sym, 32°C" = "#E64B35"
    )
  ) +
  scale_linetype_manual(
    values = c(

```

```

"Apo, 25°C" = "dashed",
"Sym, 25°C" = "solid",
"Apo, 32°C" = "dashed",
"Sym, 32°C" = "solid"
)
) +
labs(
  color = "Treatment",
  linetype = "Treatment"
) +
theme(
  legend.position = c(0.73, 0.45),
  legend.justification = c("center", "center"),
  legend.key.width = unit(2.8, "cm")
)
fig1_fuller

```



```

ggsave(
  filename = "Fig2.png",
  plot = fig1_fuller,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 7,
  height = 5,
)

```

```

units = "in",
dpi = 600,
bg = "white"
)

ggsave(
  filename = "Fig2.pdf",
  plot = fig1_fuller,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "pdf",
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)

## 2022 mortality figure

df_mortality_2022 <- tentcount_2022_cleaned %>%
  filter(
    line == "H2",
    day_cat %in% c("day_14", "day_21")
  ) %>%
  mutate(
    Mortality = ifelse(is.na(tent_count) | tent_count == 0, "Dead", "Alive"),
    dead = ifelse(Mortality == "Dead", 1, 0),
    Mortality = factor(Mortality, levels = c("Alive", "Dead")),
    day_cat = factor(day_cat, levels = c("day_14", "day_21"))
  )

my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

mortality_2022_plot_df_flipped <- df_mortality_2022 %>%
  mutate(
    sym_state = case_when(
      grepl("APO", treatment) ~ "Aposymbiotic",
      grepl("SYM", treatment) ~ "Symbiotic"
    ),
    temp_label = case_when(
      grepl("25", treatment) ~ "25°C",
      grepl("32", treatment) ~ "32°C"
    ),
    day_label = case_when(
      day_cat == "day_14" ~ "14 dpl",
      day_cat == "day_21" ~ "21 dpl"
    )
  )

```

```

    )
  ) %>%
  group_by(sym_state, temp_label, day_label) %>%
  summarise(
    mortality_prop = mean(dead, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    survival_prop = 1 - mortality_prop,
    temp_label = factor(temp_label, levels = c("25°C", "32°C")),
    sym_state = factor(sym_state, levels = c("Aposymbiotic", "Symbiotic")),
    outline_group = case_when(
      sym_state == "Aposymbiotic" & temp_label == "25°C" ~ "apo_25",
      sym_state == "Symbiotic" & temp_label == "25°C" ~ "sym_25",
      sym_state == "Aposymbiotic" & temp_label == "32°C" ~ "apo_32",
      sym_state == "Symbiotic" & temp_label == "32°C" ~ "sym_32"
    ),
    group_order = case_when(
      sym_state == "Aposymbiotic" & day_label == "14 dpl" ~ 1,
      sym_state == "Aposymbiotic" & day_label == "21 dpl" ~ 2,
      sym_state == "Symbiotic" & day_label == "14 dpl" ~ 3,
      sym_state == "Symbiotic" & day_label == "21 dpl" ~ 4
    ),
    axis_text = case_when(
      day_label == "14 dpl" ~ "14 dpl",
      day_label == "21 dpl" ~ "21 dpl"
    )
  ) %>%
  arrange(temp_label, group_order) %>%
  group_by(temp_label, sym_state) %>%
  mutate(
    bar_label = factor(axis_text, levels = rev(unique(axis_text)))
  ) %>%
  ungroup()

survival_fill_colors <- c(
  "sym_25" = "#3B6FB6",
  "apo_25" = "#6FA3D9",
  "sym_32" = "#E64B35",
  "apo_32" = "#F39B7F"
)

outline_colors <- c(
  "sym_25" = "#3B6FB6",
  "apo_25" = "#6FA3D9",
  "sym_32" = "#E64B35",
  "apo_32" = "#F39B7F"
)

p_survival_2022_base <- ggplot(
  mortality_2022_plot_df_flipped,
  aes(x = bar_label)
) +

```

```

coord_flip() +
ggh4x::facet_nested(
  rows = vars(temp_label, sym_state),
  scales = "free_y",
  space = "free_y",
  switch = "y"
) +
scale_y_continuous(
  labels = scales::percent,
  limits = c(0, 1),
  expand = expansion(mult = c(0, 0.04))
) +
labs(
  x = "Treatment Group",
  y = "Percent Survival"
) +
my_theme +
theme(
  panel.grid = element_blank(),
  strip.placement = "outside",
  strip.text.y.right = element_text(size = 14, face = "bold"),
  strip.background.y = element_rect(
    fill = "white",
    color = "black",
    linewidth = 1
  ),
  ggh4x.facet.nestline = element_blank(),
  panel.spacing.y = unit(0.08, "lines"),
  axis.text.y = element_text(size = 14, lineheight = 0.9),
  axis.text.x = element_text(size = 14),
  plot.margin = margin(8, 14, 8, 8)
)

p_survival_2022_flipped_outline <- p_survival_2022_base +
  geom_col(
    aes(y = survival_prop),
    fill = "grey80",
    width = 0.82
  ) +
  geom_col(
    aes(y = 1, color = outline_group),
    fill = NA,
    linewidth = 0.9,
    width = 0.82
  ) +
  scale_color_manual(
    values = outline_colors,
    guide = "none"
  )

p_survival_2022_flipped_colorbar <- p_survival_2022_base +
  geom_col(
    aes(y = survival_prop, fill = outline_group),

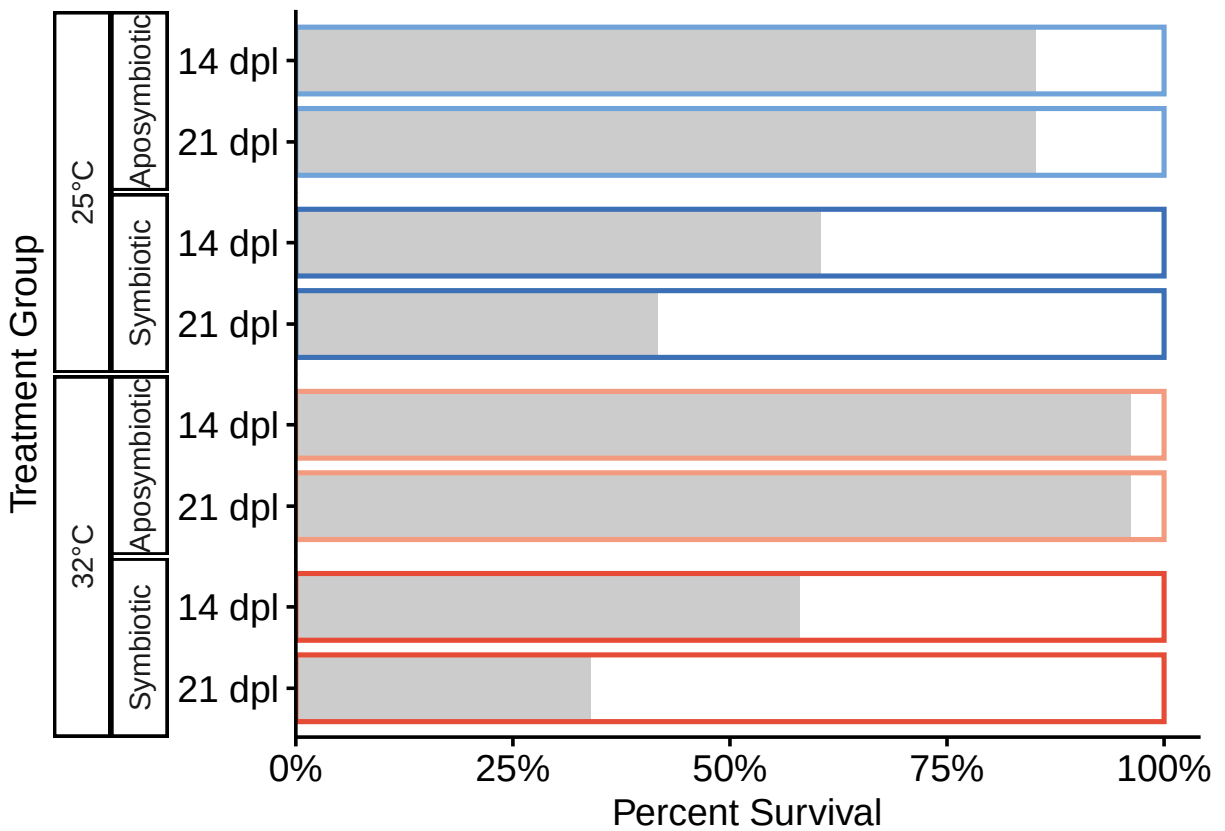
```

```

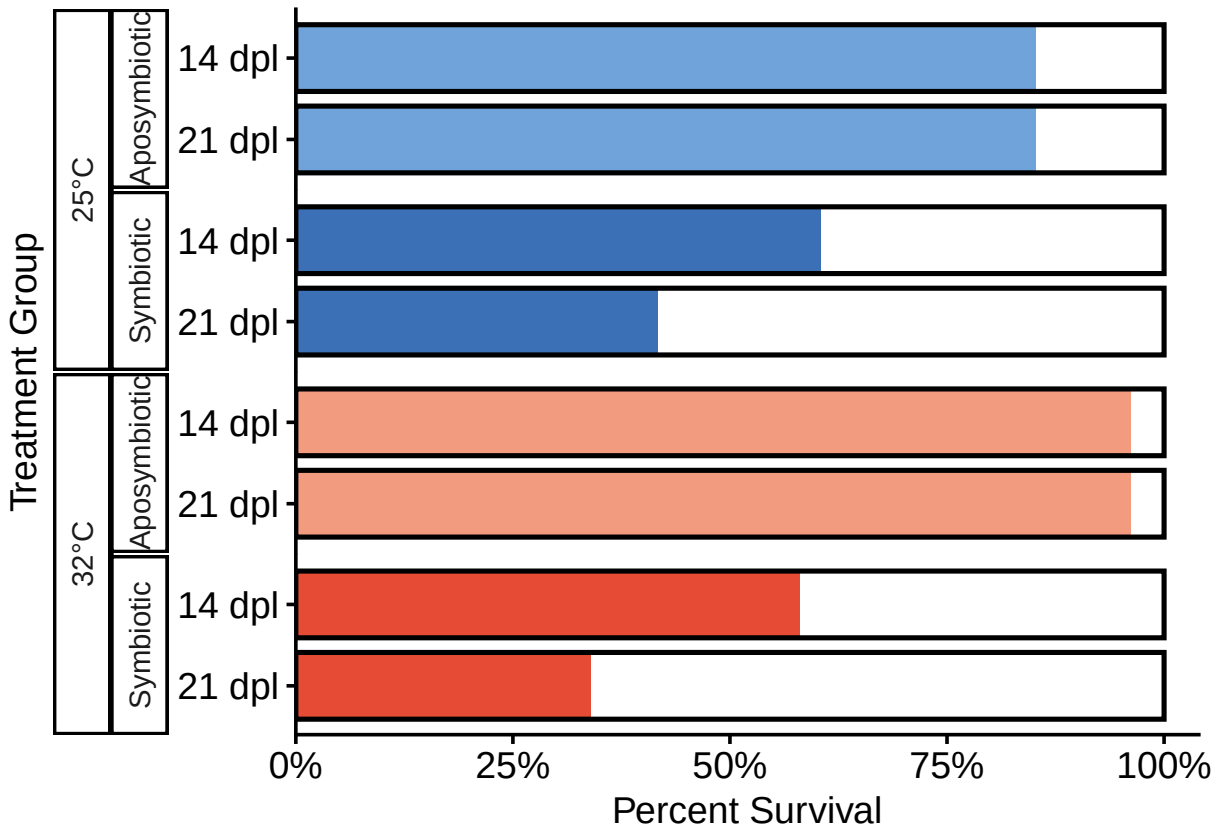
width = 0.82,
color = NA
) +
geom_col(
aes(y = 1),
width = 0.82,
fill = NA,
color = "black",
linewidth = 0.9
) +
scale_fill_manual(
values = survival_fill_colors,
guide = "none"
)

```

p_survival_2022_flipped_outline



p_survival_2022_flipped_colorbar



```

ggsave(
  filename = "Fig3_2022.png",
  plot = p_survival_2022_flipped_colorbar,
  path = "figs",
  device = "png",
  width = 7,
  height = 5,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "Fig3_2022.pdf",
  plot = p_survival_2022_flipped_colorbar,
  path = "figs",
  device = pdf,
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)

## 2022 survival stats

survival_stats_2022 <- df_mortality_2022 %>%

```

```

mutate(
  alive = 1 - dead,
  sym_state = case_when(
    grepl("APO", treatment) ~ "Aposymbiotic",
    grepl("SYM", treatment) ~ "Symbiotic"
  ),
  temp_label = case_when(
    grepl("25", treatment) ~ "25°C",
    grepl("32", treatment) ~ "32°C"
  )
) %>%
filter(
  !is.na(sym_state),
  !is.na(temp_label)
) %>%
mutate(
  sym_state = factor(
    sym_state,
    levels = c("Aposymbiotic", "Symbiotic")
  ),
  temp_label = factor(
    temp_label,
    levels = c("25°C", "32°C")
  )
)

survival_glm_2022 <- glm(
  alive ~ temp_label * sym_state,
  data = survival_stats_2022,
  family = binomial
)

summary(survival_glm_2022)

```

```

##
## Call:
## glm(formula = alive ~ temp_label * sym_state, family = binomial,
##      data = survival_stats_2022)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.7492     0.3831   4.566 4.96e-06 ***
## temp_label32°C      1.4697     0.8163   1.800  0.0718 .
## sym_stateSymbiotic  -1.7075     0.4341  -3.934 8.37e-05 ***
## temp_label32°C:sym_stateSymbiotic -1.6717     0.8651  -1.932  0.0533 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 397.21  on 301  degrees of freedom
## Residual deviance: 333.29  on 298  degrees of freedom
## AIC: 341.29

```

```
##
## Number of Fisher Scoring iterations: 5

table_survival_2022 <- summary(survival_glm_2022)$coefficients %>%
  as.data.frame() %>%
  tibble::rownames_to_column("Term")

colnames(table_survival_2022) <- c(
  "Term",
  "Estimate",
  "SE",
  "z",
  "p"
)

write.csv(
  table_survival_2022,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS3_survival_glm_2022.csv"
  ),
  row.names = FALSE
)

emm_surv_temp_2022 <- emmeans(
  survival_glm_2022,
  ~ temp_label | sym_state
)

emm_surv_sym_2022 <- emmeans(
  survival_glm_2022,
  ~ sym_state | temp_label
)

emm_surv_temp_2022
```

```
## sym_state = Aposymbiotic:
## temp_label  emmean    SE  df asymp.LCL asymp.UCL
## 25°C        1.7492 0.383 Inf    0.998    2.500
## 32°C        3.2189 0.721 Inf    1.806    4.632
##
## sym_state = Symbiotic:
## temp_label  emmean    SE  df asymp.LCL asymp.UCL
## 25°C        0.0417 0.204 Inf   -0.358    0.442
## 32°C       -0.1603 0.201 Inf   -0.554    0.233
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
emm_surv_sym_2022
```

```
## temp_label = 25°C:
## sym_state      emmean    SE  df asymp.LCL asymp.UCL
```

```
## Aposymbiotic 1.7492 0.383 Inf 0.998 2.500
## Symbiotic 0.0417 0.204 Inf -0.358 0.442
##
## temp_label = 32°C:
## sym_state emmean SE df asymp.LCL asymp.UCL
## Aposymbiotic 3.2189 0.721 Inf 1.806 4.632
## Symbiotic -0.1603 0.201 Inf -0.554 0.233
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

Supplementary heat ramp

```
library(ggplot2)
library(plotrix)
library(dplyr)
library(janitor)
library(here)
library(ggh4x)

graphics.off()

setwd("~/Documents/GitHub/heating_lacerates_final")

supp_heat_ramp_long_collapsed <- read.csv("~/Documents/GitHub/heating_lacerates_final/data/heat_ramp.csv")

str(supp_heat_ramp_long_collapsed)
```

```
## 'data.frame': 3925 obs. of 10 variables:
## $ ID : chr "H2-SYM-1_A1_X3A" "H2-SYM-1_A1_X3A" "H2-SYM-1_A1_X3A" "H2-SYM-1_A1_X3A" ...
## $ plate : chr "H2-Sym-1" "H2-Sym-1" "H2-Sym-1" "H2-Sym-1" ...
## $ treatment : chr "H2-SYM-25C (a)" "H2-SYM-25C (a)" "H2-SYM-25C (a)" "H2-SYM-25C (a)" ...
## $ line : chr "H2" "H2" "H2" "H2" ...
## $ symbiosis : chr "Sym" "Sym" "Sym" "Sym" ...
## $ well : chr "A1" "A1" "A1" "A1" ...
## $ temp : chr "25C (ambient)" "25C (ambient)" "25C (ambient)" "25C (ambient)" ...
## $ day : int 0 1 2 3 4 5 6 7 8 9 ...
## $ tent_count: int 0 0 0 0 0 0 0 0 0 0 ...
## $ day_cat : chr "day_0" "day_1" "day_2" "day_3" ...
```

```
supp_heat_ramp_long_collapsed$tent_count <- as.numeric(supp_heat_ramp_long_collapsed$tent_count)
supp_heat_ramp_long_collapsed$ID <- as.factor(supp_heat_ramp_long_collapsed$ID)
supp_heat_ramp_long_collapsed$plate <- as.factor(supp_heat_ramp_long_collapsed$plate)
supp_heat_ramp_long_collapsed$well <- as.factor(supp_heat_ramp_long_collapsed$well)
supp_heat_ramp_long_collapsed$line <- as.factor(supp_heat_ramp_long_collapsed$line)
supp_heat_ramp_long_collapsed$temp <- as.factor(supp_heat_ramp_long_collapsed$temp)
supp_heat_ramp_long_collapsed$treatment <- as.character(supp_heat_ramp_long_collapsed$treatment)
supp_heat_ramp_long_collapsed$symbiosis <- as.factor(supp_heat_ramp_long_collapsed$symbiosis)
supp_heat_ramp_long_collapsed$day <- as.numeric(supp_heat_ramp_long_collapsed$day)
supp_heat_ramp_long_collapsed$day_cat <- as.factor(supp_heat_ramp_long_collapsed$day_cat)

supp_heat_ramp_data <- supp_heat_ramp_long_collapsed %>%
  filter(
```

```

    treatment %in% c(
      "H2-SYM-25C",
      "H2-SYM-25C (a)",
      "H2-SYM-25C (b)",
      "H2-SYM-25C (c)",
      "H2-SYM-32C",
      "H2-SYM-33.5C",
      "H2-SYM-35C"
    ),
    day <= 14
  ) %>%
  mutate(
    temp = gsub("H2-SYM-", "", treatment),
    temp = gsub(" \\(.*\\)", "", temp),
    temp = factor(
      temp,
      levels = c("25C", "32C", "33.5C", "35C")
    )
  )

supp_heat_ramp_counts_rows <- supp_heat_ramp_data %>%
  count(temp, name = "n_rows")

print(supp_heat_ramp_counts_rows)

```

```

##      temp n_rows
## 1    25C   1800
## 2    32C    750
## 3  33.5C    540
## 4    35C    540

```

```

supp_heat_ramp_counts_ids <- supp_heat_ramp_data %>%
  distinct(ID, temp) %>%
  count(temp, name = "n_ids")

print(supp_heat_ramp_counts_ids)

```

```

##      temp n_ids
## 1    25C    120
## 2    32C     50
## 3  33.5C     36
## 4    35C     36

```

```

supp_heat_ramp_fig_data <- supp_heat_ramp_data %>%
  group_by(temp, day) %>%
  summarise(
    mean = mean(tent_count, na.rm = TRUE),
    se = std.error(tent_count, na.rm = TRUE),
    .groups = "drop"
  )

print(supp_heat_ramp_fig_data, n = Inf)

```

```
## # A tibble: 60 x 4
##   temp    day    mean    se
##   <fct> <dbl> <dbl> <dbl>
## 1 25C      0 0      0
## 2 25C      1 0.0167 0.0167
## 3 25C      2 0.0420 0.0302
## 4 25C      3 0.0424 0.0304
## 5 25C      4 0.0631 0.0410
## 6 25C      5 0.518  0.115
## 7 25C      6 1.48   0.193
## 8 25C      7 3.66   0.282
## 9 25C      8 5.09   0.313
## 10 25C     9 6.10   0.319
## 11 25C    10 7.09   0.292
## 12 25C    11 7.67   0.301
## 13 25C    12 8.13   0.300
## 14 25C    13 8.63   0.291
## 15 25C    14 8.70   0.290
## 16 32C      0 0      0
## 17 32C      1 0      0
## 18 32C      2 0      0
## 19 32C      3 0      0
## 20 32C      4 0.0732 0.0732
## 21 32C      5 2.27   0.353
## 22 32C      6 4      0.446
## 23 32C      7 7.20   0.480
## 24 32C      8 7.31   0.553
## 25 32C      9 7.69   0.564
## 26 32C     10 8.31   0.439
## 27 32C     11 8.31   0.439
## 28 32C     12 8.90   0.303
## 29 32C     13 9      0.314
## 30 32C     14 9      0.314
## 31 33.5C    0 0      0
## 32 33.5C    1 0      0
## 33 33.5C    2 0      0
## 34 33.5C    3 0      0
## 35 33.5C    4 0.690  0.249
## 36 33.5C    5 3.5     0.608
## 37 33.5C    6 5      0.765
## 38 33.5C    7 6.29   0.784
## 39 33.5C    8 7.39   0.755
## 40 33.5C    9 7.5     0.751
## 41 33.5C   10 8      0.642
## 42 33.5C   11 7.81   0.653
## 43 33.5C   12 7.93   0.781
## 44 33.5C   13 8.36   0.520
## 45 33.5C   14 8.5     0.584
## 46 35C      0 0      0
## 47 35C      1 0      0
## 48 35C      2 0      0
## 49 35C      3 0      0
## 50 35C      4 0.0588 0.0588
## 51 35C      5 0.0606 0.0606
```

```
## 52 35C      6 0      0
## 53 35C      7 0      0
## 54 35C      8 0      0
## 55 35C      9 0      0
## 56 35C     10 0      0
## 57 35C     11 0      0
## 58 35C     12 0      0
## 59 35C     13 0      0
## 60 35C     14 0      0
```

```
my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

supp_heat_ramp_fig <- ggplot(
  supp_heat_ramp_fig_data,
  aes(
    x = day,
    y = mean,
    color = temp,
    linetype = temp,
    shape = temp,
    group = temp
  )
) +
  my_theme +
  geom_line(linewidth = 0.9) +
  geom_errorbar(
    aes(ymin = mean - se, ymax = mean + se),
    width = 0.2,
    linewidth = 0.6
  ) +
  geom_point(size = 4) +
  ylab("Mean tentacle number") +
  xlab("Days post laceration (dpl)") +
  scale_y_continuous(
    breaks = seq(0, 15, 2),
    limits = c(0, 10)
  ) +
  scale_x_continuous(
    breaks = seq(min(supp_heat_ramp_fig_data$day), max(supp_heat_ramp_fig_data$day), 1)
  ) +
  scale_color_manual(
    values = c(
      "25C" = "#3B6FB6",
      "32C" = "#E64B35",
      "33.5C" = "#A50F15",
      "35C" = "#3B0000"
```

```

),
labels = c(
  "25C" = "Sym 25°C",
  "32C" = "Sym 32°C",
  "33.5C" = "Sym 33.5°C",
  "35C" = "Sym 35°C"
)
) +
scale_linetype_manual(
  values = c(
    "25C" = "solid",
    "32C" = "solid",
    "33.5C" = "dotted",
    "35C" = "solid"
  ),
  labels = c(
    "25C" = "Sym 25°C",
    "32C" = "Sym 32°C",
    "33.5C" = "Sym 33.5°C",
    "35C" = "Sym 35°C"
  )
) +
scale_shape_manual(
  values = c(
    "25C" = 16,
    "32C" = 16,
    "33.5C" = 17,
    "35C" = 15
  ),
  labels = c(
    "25C" = "Sym 25°C",
    "32C" = "Sym 32°C",
    "33.5C" = "Sym 33.5°C",
    "35C" = "Sym 35°C"
  )
) +
labs(
  color = "Temperature",
  linetype = "Temperature",
  shape = "Temperature"
) +
guides(
  color = guide_legend(
    override.aes = list(
      linewidth = 1.5,
      linetype = c("solid", "solid", "dotted", "solid"),
      shape = c(16, 16, 17, 15)
    )
  ),
  linetype = "none",
  shape = "none"
) +

```



```

theme(
  legend.position = c(0.79, 0.43),
  legend.justification = c("center", "center"),
  legend.key.width = unit(1.6, "cm")
)

supp_heat_ramp_fig

ggsave(
  filename = "FigS1_heat_ramp.png",
  plot = supp_heat_ramp_fig,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 7,
  height = 5,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "FigS1_heat_ramp.pdf",
  plot = supp_heat_ramp_fig,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "pdf",
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)

treatment_cols <- c(
  "apo_25" = "#6FA3D9",
  "inoc_25" = "#3B88C3",
  "sym_25" = "#3B6FB6",
  "apo_32" = "#F39B7F",
  "inoc_32" = "#D95F02",
  "sym_32" = "#E64B35"
)

```